TCC805x MCU BSP

API Specification for Safe Display Manager

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the safe display manager application driver (here in after referred to as SDM app driver) of TCC805x MCU BSP. The SDM app driver shows how to control the SDM device using GPSB. For more detailed information of the safe display manager in the TCC805x MCU Sub-system, refer to the *"PART 12-SDM SUB-SYSTEM"* in the *"TCC805x Full Specification"*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

SDM	Safe Display Manager
TM	Timing Manager in SDM
GPSB	General Purpose Serial Bus

3 SOURCE FILES

3.1 Location of Source Files

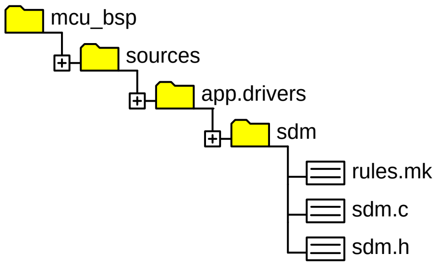

Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- rules.mk
  - SDM building rules
- sdm.c
  - SDM source file. Sets SDM by using GPSB and default screen output.
- sdm.h
  - SDM header file

4 ENUMERATIONS

4.1 SDMError_t

Enumeration variables for SDM error return type.

Declaration

```
typedef enum
{
    SDM_ERR_NONE              = 0,
    SDM_ERR_FAIL              = 1,
    SDM_ERR_INVALID_PARAM     = 2,
    SDM_ERR_SPI_NOT_INIT      = 3,
    SDM_ERR_XFER_REG          = 4,
} SDMError_t;
```

Description

Value	Description
SDM_ERR_NONE	No error
SDM_ERR_FAIL	Common error
SDM_ERR_INVALID_PARAM	When entering an invalid parameter, this error is returned.
SDM_ERR_SPI_NOT_INIT	GPSB access without opening GPSB
SDM_ERR_XFER_REG	GPSB transfer error

Remark

None

5 CONSTANTS

5.1 SDM Register Address

Definition variables for SDM register address.

Declaration

```
#define SDM_VA_CLK_ADDR       (0x0UL)
#define SDM_VA_CONF_ADDR      (0x100000UL)
#define SDM_VA_VS_ADDR        (0x820000UL)
#define SDM_VA_TT_DATA_ADDR   (0x800000UL)
#define SDM_VA_TT_HEAD_ADDR   (0x820200UL)
#define SDM_VA_AS_ADDR        (0x900000UL)
```

Description

Value	Description
SDM_VA_CLK_ADDR	SDM CKC Register address. (This address can only be accessed using GPSB.)
SDM_VA_CONF_ADDR	SDM Configuration Register address. (This address can only be accessed using GPSB.)
SDM_VA_VS_ADDR	SDM Video supervisor Register address. (This address can only be accessed using GPSB.)
SDM_VA_TT_DATA_ADDR	Telltale memory address to store compressed telltale data. The maximum size is 600kb. (This address can only be accessed using GPSB.)
SDM_VA_TT_HEAD_ADDR	SDM Video supervisor register 2 address for telltale header. (This address can only be accessed using GPSB.)
SDM_VA_AS_ADDR	SDM Audio supervisor register address. (This address can only be accessed using GPSB.)

Remark

None

6 APIs

6.1 SDM_ReadReg

Reads a value of SDM register by using GPSB driver.

Declaration

```
uint8 SDM_ReadReg
(
    uint32      uiAddr,
    uint32 *    piValue
);
```

Parameters

Value	Description
uiAddr	SDM register address
piValue	SDM register value

Return

Value	Description
SDM_ERR_SPI_NOT_INIT	Need to open GPSB device using **SDM_Init** function
SDM_ERR_INVALID_PARAM	Invalid parameter
SDM_ERR_XFER_REG	GPSB transmission failure
SDM_ERR_NONE	Register read done

Remark

The second parameter, ***piValue***, gets 32 bits from the register.

6.2 SDM_WriteReg

Writes a value of SDM register by using GPSB driver.

Declaration

```
uint8 SDM_WriteReg
(
    uint32          uiAddr,
    const uint8 *   pcValue,
    uint32          uiSize
);
```

Parameters

Value	***Description***
uiAddr	*SDM register address*
pcValue	*SDM register value*
uiSize	*Write size*

Return

Value	***Description***
SDM_ERR_SPI_NOT_INIT	*Need to open GPSB device using **SDM_Init** function*
SDM_ERR_INVALID_PARAM	*Invalid parameter*
SDM_ERR_XFER_REG	*GPSB transmission failure*
SDM_ERR_NONE	*Register read done*

Remark

The minimum writing unit is 32 bits. The second parameter, ***pcValue***, sets data by 8 bits in this register. If you want to write to the SDM register, you need at least a 32-bit array of 8 bits.

6.3 SDM_Init

This function is called in ***main.c*** and displays the video data through ***LCD_MUX2***. This function shows how to access the SDM register through GPSB. When outputting a screen through SDM, the settings in this function are absolutely necessary. This function sets width and height, which are the default for screen output. This function also sets VIDEO MUX to DDI to receive video data.

Declaration

```
uint8 SDM_Init
(
    void
);
```

Parameters

None

Return

Value	*Description*
SDM_ERR_SPI_NOT_INIT	*Need to open GPSB device using **SDM_Init** function*
SDM_ERR_INVALID_PARAM	*Invalid parameter*
SDM_ERR_XFER_REG	*GPSB transmission failure*
SDM_ERR_NONE	*Register read done*

Remark

In order to display the screen through SDM, AP core must transmit video data to***LCD_MUX2*** as shown below.

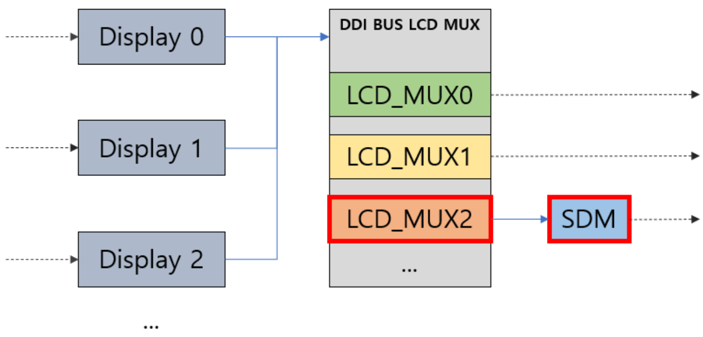

Figure 2 DDI LCD_MUX Setting

For information on LCD_MUX setting, refer to the *"**TCC805x Automotive Common BSP-Application Note for U-boot Display Output Configuration**"*.

7 How to use SDM Driver

7.1 Initialize SDM app driver

SDM_Init() writes an appropriate screen resolution to the TM_CTRL register, to output the data received from ***LCD_MUX2*** to the display device. This function performs the most basic setting of displaying a screen on a display device using SDM. Default resolution is set to 1920x720.

```
{
    SDMError_t ucRet;
    ucRet = SDM_Init();
}
```

7.2 Write SDM Register

```
{
    const uint8    ucSdmUnLock[4] = {0x1A, 0xCC, 0xE5,0x51};
    const uint8    ucSdmlock[4]   = {0xFF, 0xFF, 0xFF,0xFF};
    const uint8    ucVa[4]        = {0,0,0,2};                /* DDI sel */
    const uint8    ucVaTmCtl4[4]  = {0x07,0x80,0x00,0x00};    /* w:1920 */
    const uint8    ucVaTmCtl6[4]  = {0x02,0xD0,0x00,0x00};    /* h:720 */
    SDMError_t     ucRet;

    ucRet   = SDM_WriteReg(SDM_VA_VS_ADDR, ucSdmUnLock, 4UL);
    ucRet   = SDM_WriteReg(SDM_VA_VS_ADDR + 0x40UL, (const uint8*)ucVaTmCtl4, 4UL);
    ucRet   = SDM_WriteReg(SDM_VA_VS_ADDR + 0x48UL, (const uint8*)ucVaTmCtl6, 4UL);
    ucRet   = SDM_WriteReg(SDM_VA_CONF_ADDR + 0x4CUL, (const uint8*)ucVa, 4UL);
    ucRet   = SDM_WriteReg(SDM_VA_VS_ADDR, ucSdmLock, 4UL);
}
```

7.3 Read SDM Register

```
{
    SDMError_t ucRet;
    uint32 uiReadVa;

    uiReadVa   = 0;
    ucRet      = SDM_ReadReg(SDM_VA_CONF_ADDR, &uiReadVa);
    printf(“Read : 0x%08x\n”, uiReadVa);
}
```

8 REFERENCES

[1] TCC805x Full Specification

[2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release